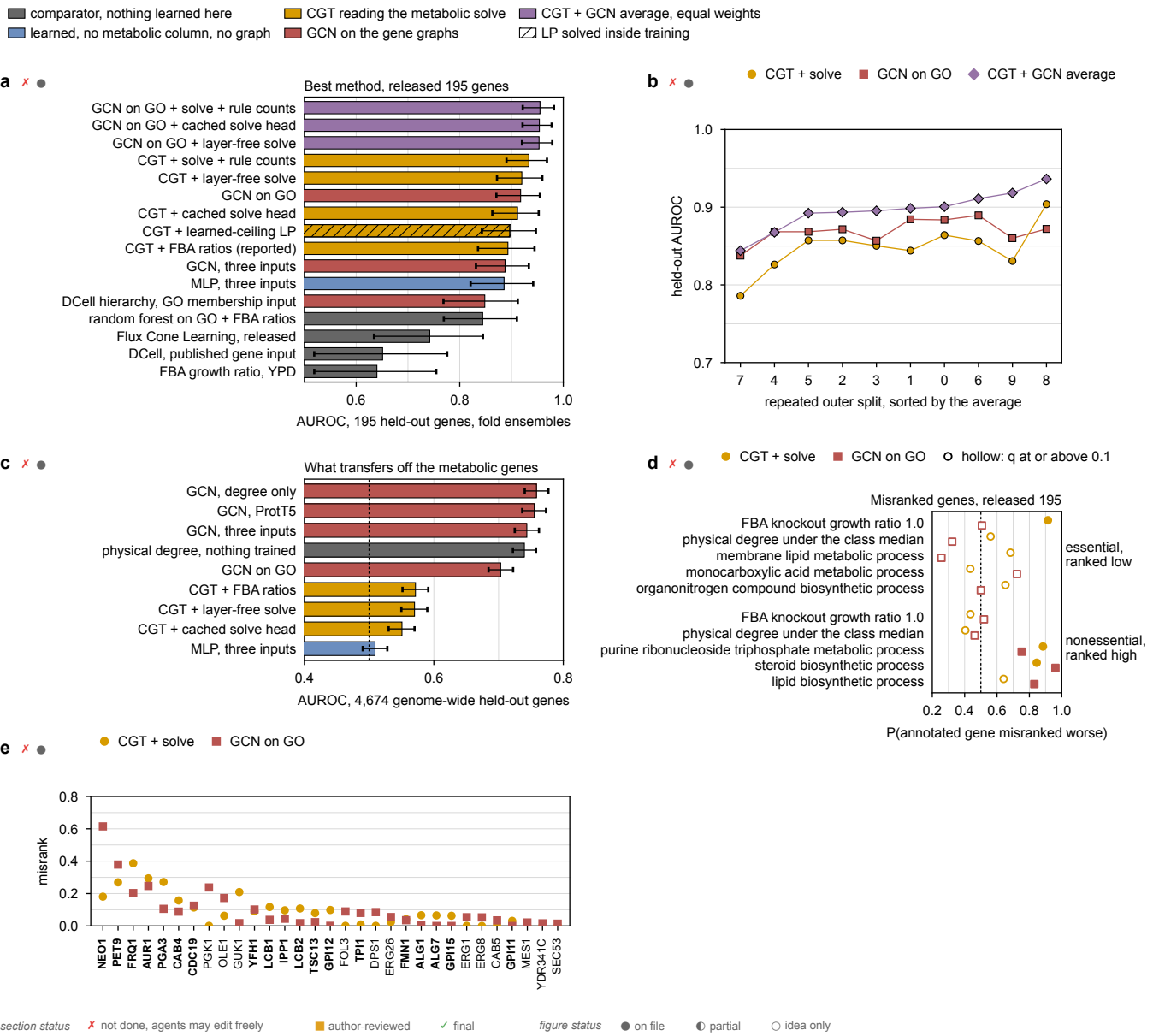

comparator, nothing learned here
learned, no metabolic column, no graph
CGT reading the metabolic solve
GCN on the gene graphs
CGT + GCN average, equal weights
LP solved inside training
a
Best method, released 195 genes
GCN on GO + solve + rule counts
GCN on GO + cached solve head
GCN on GO + layer-free solve
CGT + solve + rule counts
CGT + layer-free solve
GCN on GO
CGT + cached solve head
CGT + learned-ceiling LP
CGT + FBA ratios (reported)
GCN, three inputs
MLP, three inputs
DCell hierarchy, GO membership input
random forest on GO + FBA ratios
Flux Cone Learning, released
DCell, published gene input
FBA growth ratio, YPD
0.6
0.8
1.0
AUROC, 195 held-out genes, fold ensembles
b
CGT + solve
GCN on GO
CGT + GCN average
held-out AUROC
1.0
0.9
0.8
0.7
7 4 5 2 3 1 0 6 9 8
repeated outer split, sorted by the average
c
What transfers off the metabolic genes
GCN, degree only
GCN, ProtT5
GCN, three inputs
physical degree, nothing trained
GCN on GO
CGT + FBA ratios
CGT + layer-free solve
CGT + cached solve head
MLP, three inputs
0.4
0.6
0.8
AUROC, 4,674 genome-wide held-out genes
d
CGT + solve
GCN on GO
hollow: q at or above 0.1
Misranked genes, released 195
FBA knockout growth ratio 1.0
physical degree under the class median
membrane lipid metabolic process
monocarboxylic acid metabolic process
organonitrogen compound biosynthetic process
FBA knockout growth ratio 1.0
physical degree under the class median
purine ribonucleoside triphosphate metabolic process
steroid biosynthetic process
lipid biosynthetic process
essential, ranked low
nonessential, ranked high
0.2 0.4 0.6 0.8 1.0
P(annotated gene misranked worse)
e
CGT + solve
GCN on GO
misrank
0.8
0.6
0.4
0.2
0.0
NEO1 PET9 FRQ1 AUR1 PGA3 CAB4 CDC19 PGK1 OLE1 GUK1 YFH1 LCB1 IPP1 LCB2 TSC13 GPI12 FOL3 TPI1 DPS1 ERG26 FMN1 ALG1 ALG7 GPI15 ERG1 ERG8 CAB5 GPI11 MES1 YDR341C SEC53
section status
✗ not done, agents may edit freely
author-reviewed
✓ final
figure status
on file
partial
idea only